Curriculum Vitae

Wenli Zhao
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Research Profile

Climate-energy-environment interactions and urban resilience; observation-constrained hybrid Earth-system modeling; ecohydrology and hydroclimate extremes; remote sensing; physics-guided and explainable machine learning; power-system carbon analytics.

Academic Appointments

Spring 2026 Adjunct Assistant Professor, Columbia University, New York, NY
Instructor of record, Hydrosystems Engineering (EACE E3250/E4250), Spring 2026

2025-present Associate Research Scientist, Columbia University, New York, NY

2023-2025 Postdoctoral Research Scientist, Columbia University, New York, NY
Supervisor: Pierre Gentine

2022-2023 Postdoctoral Research Scientist, Max Planck Institute for Biogeochemistry, Jena, Germany
Supervisors: Markus Reichstein and Alexander Winkler

2020-2022 Postdoctoral Research Scientist, Columbia University, New York, NY
Supervisor: Pierre Gentine

Education

2015-2020 Ph.D., Environmental Science, Peking University, Beijing, China
Dissertation: Physics-Constrained Machine Learning of Evapotranspiration; advisors: Guoyu Qiu and Pierre Gentine

2011-2015 B.E., Geographic Information System, Northeast Normal University, Changchun, China
Advisor: Shan Lu

Research Support

Current Award

2025-2028 NASA ROSES-2024 A.63 Ecohydrology, Grant 80NSSC26K0079
Enhancing Mechanistic Understanding of Vegetation-Environment Interactions During Pulse Events with a Differentiable Land Model and Observations | Total award: $694,276 | Role: Institutional PI/Co-Investigator | Led scientific proposal development and writing; selected as one of 3 awards from 46 submissions

Publications and Manuscripts

*# Co-first author; * corresponding author; † student-led work advised or co-advised by Wenli Zhao, with contributions to study conception and design. Manuscripts under review or revision are listed separately from published papers.*

First- or Corresponding-Author Peer-Reviewed Publications

Zhao, W.*, Winkler, A. J., Reichstein, M., Orth, R., & Gentine, P. (2026). Learning Evaporative Fraction with Memory. Hydrology and Earth System Sciences. In production; <u>Highlight Paper</u>. Preprint: https://doi.org/10.5194/egusphere-2025-4082 (>4,500 PDF downloads since August 2025).

Zhao, W.*, et al. (2026). Observation-constrained physical snow water equivalent simulations using a physics-guided machine learning approach. Water Resources Research, 62, e2025WR041406. https://doi.org/10.1029/2025WR041406

Zhang, S.#†, **Zhao, W.**#*, et al. (2025). A near-real time daily European Power Consumption and Carbon Intensity Dataset (ECON-PowerCI). Scientific Data, 12, 1693. https://doi.org/10.1038/s41597-025-05978-7

Shi, Z.†, **Zhao, W.***, et al. (2026). A theoretical framework for critical canopy temperature and its application to improve remote-sensing-based estimation of evapotranspiration. Environmental Research Letters, 21, 024011. https://doi.org/10.1088/1748-9326/ae33d3

Zhao, W.*, Zhu, B., Davis, S. J., Ciais, P., et al. (2023). Reliance on fossil fuels increases during extreme temperature events in the continental United States. Communications Earth & Environment, 4, 473. https://doi.org/10.1038/s43247-023-01147-z. <u>Top 94th percentile among all research outputs tracked by Altmetric.</u>

Zhao, W., Qiu, G. Y., Xiong, Y. J., Gentine, P., & Chen, B. Y. (2020). Uncertainties caused by resistances in evapotranspiration estimation using high-density eddy covariance measurements. Journal of Hydrometeorology, 21, 1349-1365. https://doi.org/10.1175/JHM-D-19-0191.1

Zhao, W.*, Gentine, P., Reichstein, M., et al. (2019). Physics-constrained machine learning of evapotranspiration. Geophysical Research Letters, 46, 14496-14507. https://doi.org/10.1029/2019GL085291. Recognized as an <u>ESI Highly Cited Paper</u> (top 1% in Geosciences).

Manuscripts Under Review or Revision

Zhao, W.*, Zhang, S., Zhu, B., et al. (2026). A near-real-time daily European dataset of fuel-specific consumption-based power carbon intensity. Under review.

Xiong, Y., Jiang, J., Blougouras, G., ... **Zhao, W.*** (2026). Higher vegetation thermal thresholds through adaptation sustain urban cooling. Under review.

Jiang, J.#, **Zhao, W.#**, Xiao, Z., Yan, C., Mo, D., Jiang, S., & Xiong, Y.* (2026). Estimating daily 10-m land surface temperature by integrating satellite embeddings with physics-guided deep learning. Under review.

Jiang, J.#, **Zhao, W.#**, Yan, C., Qiu, G. Y., et al. (2026). LoRAET: Natural-to-urban adaptation for daily urban evapotranspiration mapping at 10-m resolution. Under review.

Zhao, W.*, Zhang, S., Fang, J., et al. (2026). Multi-Scale Memory Governs Ground Heat Flux Dynamics. Under review. Preprint: https://doi.org/10.2139/ssrn.6852378

Ning, X.†, **Zhao, W.***, et al. (2025). Unveiling the carbon-emission mechanism of the global power system under extreme weather: Country-specific insights through explainable machine learning. Under review. Preprint: https://doi.org/10.2139/ssrn.5405302

Cattry, M., **Zhao, W.**, Nathaniel, J., Qiu, J., Zhang, Y., & Gentine, P. (2025). EcoPro-LSTM v0: A memory-based machine-learning approach to predicting ecosystem dynamics across time scales in Mediterranean environments. Under review. Preprint: https://doi.org/10.5194/egusphere-2024-3726

Other Peer-Reviewed Publications

Fang, J.†, Bowman, K., **Zhao, W.**, Lian, X., & Gentine, P. (2026). Differentiable land model reveals global environmental controls on latent ecological functions. Nature Communications, 17, 6670. https://doi.org/10.1038/s41467-026-73395-4

Huang, F., Jiang, S., Shangguan, W., Camps-Valls, G., Winkler, A. J., Li, W., Duveiller, G., Reimers, C., **Zhao, W.**, Reichstein, M., & Dai, Y. (2026). Upwind terrestrial influences on soil moisture variability across South America. Nature Communications, 17, 7256. https://doi.org/10.1038/s41467-026-75637-x

Jiang, H., Yao, L., Qin, J., **Zhao, W.**, et al. (2026). Building façade photovoltaics enhance global climate resilience. Nature Climate Change, 16, 566-574. https://doi.org/10.1038/s41558-026-02606-z

Jiang, H., Yao, L., Qin, J., Bai, Y., Brandt, M., Lian, X., Davis, S. J., Lu, N., **Zhao, W.**, Liu, T., & Zhou, C. (2025). Globally interconnected solar-wind system addresses future electricity demands. Nature Communications, 16, 4523. https://doi.org/10.1038/s41467-025-59879-9

Worden, M. A., Bilir, T. E., Bloom, A. A., ... **Zhao, W.**, & Zhu, S. (2025). Combining observations and models: A review of the CARDAMOM framework for data-constrained terrestrial ecosystem modeling. Global Change Biology, 31(8), e70462. https://doi.org/10.1111/gcb.70462

Yan, W., Jiang, J., He, L., **Zhao, W.**, et al. (2024). Correcting land surface temperature from thermal imager by considering heterogeneous emissivity. International Journal of Applied Earth Observation and Geoinformation, 129, 103824.

Zhu, B., Deng, Z., Song, X., **Zhao, W.**, et al. (2023). CarbonMonitor-Power: Near-real-time monitoring of global power generation on hourly to daily scales. Scientific Data, 10, 217. https://doi.org/10.1038/s41597-023-02094-2

Lian, X., **Zhao, W.**, & Gentine, P. (2022). Recent global decline in rainfall interception loss due to altered rainfall regimes. Nature Communications, 13, 7642. https://doi.org/10.1038/s41467-022-35414-y

Yu, L., Qiu, G., Yan, C., **Zhao, W.**, et al. (2022). A global terrestrial evapotranspiration product based on the three-temperature model with fewer input parameters and no calibration requirement. Earth System Science Data, 14, 3673-3693. https://doi.org/10.5194/essd-14-3673-2022

Mao, P., Qin, L., Hao, M., **Zhao, W.**, et al. (2021). An improved approach to estimate above-ground volume and biomass of desert shrub communities based on UAV RGB images. Ecological Indicators, 125, 107494.

Hao, M., **Zhao, W.**, Qin, L., et al. (2021). A methodology to determine the optimal quadrat size for desert vegetation surveying based on UAV RGB photography. International Journal of Remote Sensing, 42, 84-105.

Xiong, Y., **Zhao, W.**, Wang, P., et al. (2019). Simple and applicable method for estimating evapotranspiration and its components in arid regions. Journal of Geophysical Research: Atmospheres, 124, 9963-9982.

Yan, C., **Zhao, W.**, Wang, Y., et al. (2017). Effects of forest evapotranspiration on soil water budget and energy flux partitioning in a subalpine valley of China. Agricultural and Forest Meteorology, 246, 207-217.

Lu, S., Lu, X., **Zhao, W.**, et al. (2015). Comparing vegetation indices for remote chlorophyll measurement of white poplar and Chinese elm leaves with different adaxial and abaxial surfaces. Journal of Experimental Botany, 66, 5625-5637.

Chinese-Language Peer-Reviewed Publications

Luo, J., Qin, L., Mao, P., Xiong, Y., **Zhao, W.**, Gao, H., & Qiu, G. (2021). Research progress in retrieval algorithms for chlorophyll-a, a key element of water-quality monitoring by remote sensing. Remote Sensing Technology and Application, 36(3), 473-488. In Chinese.

Zhao, W., Xiong, Y., Qiu, G., et al. (2021). Impact of model structure and parameterization differences on evapotranspiration estimation. Acta Scientiarum Naturalium Universitatis Pekinensis, 57(1), 162-172. In Chinese.

Zhao, W., Qiu, G., Xiong, Y., et al. (2021). Simulation of sub-daily transpiration characteristics of typical urban trees based on a deep neural network. Acta Scientiarum Naturalium Universitatis Pekinensis, 57(2), 322-332. In Chinese.

Yang, Y., Zou, Z., **Zhao, W.**, & Qiu, G. (2017). Comparative study on the thermal-environment effects of six urban underlying surfaces. Acta Scientiarum Naturalium Universitatis Pekinensis, 53(5), 881-889. In Chinese.

Yang, Y., Zou, Z., **Zhao, W.**, Li, R., & Qiu, G. (2016). Experimental study on surface thermal-environment characteristics of urban pavement bricks using hyperspectral technology. Ecology and Environmental Sciences, 25(5), 835-841. In Chinese.

Yang, T., **Zhao, W.**, Wang, Z., Lu, X., & Lu, S. (2015). Changes in China's cropping system based on remotely sensed NDVI data. Scientia Agricultura Sinica, 48(10), 1915-1925. In Chinese.

Selected Presentations

2025	How Interpretable Machine Learning Advances Our Understanding of Climate Extremes and Their Impacts on Ecosystems and Infrastructure. AGU Fall Meeting. Oral presentation.
2025	Changes in the carbon cycle and its linkage to the water cycle during the 2023-2024 El Niño in the Amazon. AGU Fall Meeting. Poster.
2023	An objective estimate of water stress: Going beyond PDSI. EGU General Assembly. Poster.
2022	Extreme events increase carbon emissions and reliance on fossil fuels. AGU Fall Meeting. Poster.
2022	Physics-constrained machine learning of streamflow. HydroML Symposium. Oral presentation.
2020	Physics-constrained machine learning of canopy interception. AGU Fall Meeting. eLightning presentation.
2019	Physics-constrained machine learning of evapotranspiration. AGU Fall Meeting. Oral presentation.

Teaching and Mentoring

Spring 2026 Instructor of Record, Hydrosystems Engineering (EACE E3250/E4250), Columbia University
26 students (14 undergraduate, 12 graduate); responsible for lectures, assignments, in-class activities, and office hours

2022 Guest Lecturer, Max Planck Institute for Biogeochemistry
How Extreme Events Challenge Global Power-System Decarbonization

2021 Guest Lecturer, Peking University
Machine Learning in Environmental Science: Lessons from Hybrid Modeling of Evapotranspiration

2024-present Research mentor: Julia L. Simpson, Ph.D. candidate, Columbia University
Transfer learning for air-sea flux parameterizations

2023-present Research mentor: Jianing Fang, Ph.D. candidate, Columbia University
Differentiable land modeling and global ecological parameters

2023-present Research mentor: Ning Xuan, graduate student, Peking University
Explainable machine learning of global power-system emissions under extreme weather

2022-present Research mentor: Zhe Shi, Ph.D. candidate, Peking University
Critical canopy temperature and remote-sensing-based evapotranspiration

2022-2025 Research mentor: Shujie Zhang, graduate student, Peking University
ECON-PowerCI data development and applications

Professional Service

Editorial Board member, Chinese Geographical Science (2024-present).

Session chair, AGU 2025: High-Resolution Modeling and Untangling Atmosphere-Hydrology-Ecology Interactions.

Session chair, AGU 2025: Toward a Hybrid Era of Physical and Machine-Learning Models for Extreme Events.

Session chair, AGU 2024: Modeling and Untangling Climate-Hydrology-Ecology Interactions Through Extreme Events.

Judge, Outstanding Student and PhD Candidate Presentation (OSPP) Awards, EGU General Assembly (2023).

Reviewer for Nature Geoscience, Geophysical Research Letters, Water Resources Research, Journal of Hydrology, Hydrology and Earth System Sciences, Agricultural and Forest Meteorology, Remote Sensing of Environment, JAMES, Journal of Geophysical Research: Biogeosciences, Geoscientific Model Development, Biogeosciences, and other journals.

Honors and Awards

2022	European Commission Seal of Excellence, MSCA Postdoctoral Fellowship proposal
2015-2018	First-Class Postgraduate Scholarship, Peking University
2016-2017	Learning Excellence Award, Peking University
2015-2016	Special Award, Challenge Cup Scientific Research Competition, Peking University
2013-2015	Selected member, elite undergraduate class, Northeast Normal University
2012-2013	National Students' Innovation and Entrepreneurship Training Program

Technical Expertise

Python, MATLAB, C#, Linux, Google Earth Engine, ArcGIS, ENVI, IDL, GDAL, SQL Server, SPSS, Origin; remote sensing, geospatial analysis, deep learning, hybrid AI-physics modeling, explainable machine learning, process-based land and hydrologic modeling.